



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Instrumentation Engineering
In-Semester Examination I
Academic Year: 2025-26 (semester I)

Class: Final Year B.Tech.

Course: Elective: Building Automation System

Course Code: PEC-IN406

Date: 18.08.2025

Time: 11:00 to 11:45

Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.
- vi. Solve any four of the following.

Q.N.		Marks	CO	BT
Q.1	Define Intelligent Building in three different categories.	05	CO1	BT1
Q.2	Write a short note on Intelligent architecture and structure.	05	CO1	BT2
Q.3	What is BAS? List out its benefits and explain any one of them in detail.	05	CO1	BT4
Q.4	What do you mean by BACNet objects? Explain in detail.	05	CO2	BT4
Q.5	Explain any of the following protocol in detail, A. Profibus B. EIB	05	CO2	BT2
Q.6	Explain LonWorks protocol architecture in detail.	05	CO2	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Get the knowledge of basic fundamentals of Building Automation and management system.
CO 2	Describe BAS Communication standards, Internet technologies used in building automation such as BACnet, LonWorks, Modbus, PROFIBUS and EIB.
CO 3	Employ different kinds of systems to develop optimized way of air- conditioning, ventilation, heating, cooling and central chilling.
CO 4	Devise Lighting- control system using standard lighting control protocols, common automation protocols for energy management and lighting control so as to achieve typical benefits of BAS.
CO 5	Support building management function by providing security and safety systems to people and equipment.
CO6	Design a total building management system to provide a individual a highly efficient, comfortable, convenient and safe environment by satisfying four fundamental demands Structure, System, Service and management and optimizing their interrelationship.



S. G. G. S. Institute of Engineering and Technology, Vishnupuri, Nanded.
Internal semester Examination-I
Class: Final Year B. Tech (Instrumentation)
Code-PCC-IN403

Date: 19/08/2025
Time: 11.00 TO
11.45 am

Subject: Biomedical Instrumentation

Max. Marks : 20

Note: Solve all questions.

Sr. No	Question	Marks	COs BT
Q. 1	A. Draw block diagram of generalized biomedical instrumentation system and explain each block in detail.	03	CO1 BT CO2 1, CO3 BT2
	B. Fill in the blanks:	02	
	1. Micro shocks occur on -----.		
	2. The bioelectric generator of heart is situated at -----.		
	3. The normal PH of the blood is-----.		
Q.2	4. The blood pressure of a normal person is-----.		
	A. Discuss the development of action potential and muscular contraction.	03	CO4 BT1 CO2
	B. How does the blood circulate throughout the body? Explain it with proper diagram.	02	
Q.3	A. What are the types of surface electrodes? Explain each in detail.	03	CO2 BT1 CO1
	B. What is blood pressure? Explain using sphygmomanometer blood pressure measurement method.	02	
Q.4	A. Describe the possibilities of occurrence of electrical accidents for a patient in a hospital.	03	CO6 BT4 CO5
	B. What are the factors which should be taken into account during designing of a medical equipment?	02	

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

On successful completion of this course, students should be able to:

- CO 1 Understand biomedical instrumentation, with transducer, electrode used.
- CO 2 Study Electrographs, Physiological pressure measurements, Respiratory system.
- CO 3 Study Instrumentation for measuring Brain parameters.
- CO 4 The students will have a clear knowledge about human physiology system.
- CO 5 They will have knowledge of the principle operation and design and the background.
- CO6 Knowledge of biomedical instruments and specific applications of biomedical engineering.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.

Department of Instrumentation Engineering

In-Semester Examination I

Academic Year: 2025-26 (semester I)

Class: Final Year B.Tech.

Course: Modern Control Theory

Course Code: PCC-IN401

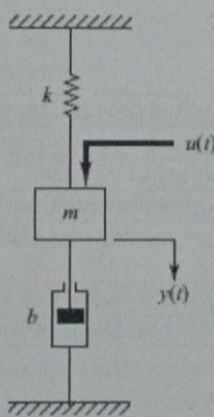
Date: 21.08.2025

Time: 11:00 to 11:45

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.
- Solve any 4 questions from following 5 questions

Q.N.		Marks	CO	BT
Q.1	Derive state space equation for following mechanical system and draw the block diagram  Where, k=spring constant, M= Mass, b=damping coefficient, $u(t)$= Applied force or input variable, $y(t)$= output variable or displacement.	5	CO1	BT3
Q.2	Derive a transfer function equation from given state space equation. $\dot{x}(t) = Ax(t) + Bu(t)$ $y(t) = cx(t) + Du(t)$ Where, $A = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ -160 & -56 & -14 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1 \\ -14 \end{bmatrix}, C = [1 \ 0 \ 0], D = [0]$	5	CO1	BT2
Q.3	Consider the following state space equation with state variable $x(t)$, find out new state variable $z(t)$ by using diagonalization of state matrix method. $\dot{x}(t) = Ax(t) + Bu(t)$	5	CO2	BT3

	$y(t) = cx(t) + Du(t)$ Where, $A = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1 \\ 6 \end{bmatrix}, C = [1 \ 0 \ 0], D = [0]$			
Q.4	Calculate state transition matrix for given state matrix. $A = \begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix}$	5	CO2	BT5
Q.5	Check the controllability and observability of following state space process model. $\dot{x}(t) = Ax(t) + Bu(t)$ $y(t) = cx(t) + Du(t)$ Where, $A = \begin{bmatrix} 1 & 1 \\ -2 & -1 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, C = [1 \ 0], D = [0]$	5	CO2	BT4

== ALL THE BEST == ✓

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Know the concept of state, state variable, state model and state space representation of physical systems.
CO 2	Illustrate stability, controllability and observability of a system.
CO 3	Apply knowledge of control theory for practical implementations in engineering and network analysis.
CO 4	Analyze dynamics of a linear system by solving system model/equation or applying domain transformation.
CO 5	Test non linearity of the system and evaluate various techniques for finding stability of nonlinear system.
CO 6	Formulate and solve deterministic optimal control problems in terms of performance indices.